

Abhijith Shaji

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EDUCATION

- 08/24 – Present **University of Southern California, Los Angeles**
PhD in Computer Science, GPA: 3.90 / 4.0
Coursework: Advanced Computer Vision, Probabilistic and Generative Models
- 07/20 – 06/24 **Indian Institute of Technology Tirupati, India**
B. Tech in Computer Science and Engineering, GPA: 9.16 / 10.0
Coursework: Linear Algebra, Computer Organization, Deep Learning

RESEARCH EXPERIENCE

- 05/25 – Present **Graduate Researcher**, Roski Eye Institute, University of Southern California.
Advisor: [Prof. Benjamin Xu](#)
- Fine-tuned the RETFound foundation model to classify fundus photographs into the diagnostic categories: Glaucoma suspects, Mild Glaucoma and Severe Glaucoma. It was then compared against the performance of Glaucoma specialists. [\[IOVS 2026\]](#)
 - Evaluated diagnostic agreement of publicly available large language models (LLMs) against Glaucoma specialists in classifying Glaucoma versus Glaucoma suspect from cross-sectional multimodal clinical data, benchmarked against consensus derived reference standards. [\[Ophthalmology Science 2026\]](#)
 - Built multi-modal neural networks integrating Vision Transformer (ViT) image embeddings with ANN-derived structured data embeddings to enhance the performance of the existing Glaucoma detection algorithms.
- 08/24 – Present **Graduate Researcher**, Information Sciences Institute and Computer Science Department, University of Southern California.
Advisor: [Prof. Jose-Luis Ambite](#)
- Working on conditional generative latent diffusion models (LDMs) in a federated learning setting to study the quality of synthetic diffusion-weighted brain images generated using a secure federated learning architecture.
 - Designed conditional generative diffusion models to synthesize missing brain MRI modalities from available modalities, and evaluated the performance of multimodal 3D CNNs incorporating imputed MRI scans for Alzheimer's disease (AD) classification, compared against unimodal 3D CNN baselines. [\[IEEE EMBC 2026\]](#)
- 08/23 – 03/24 **Independent Researcher** (B. Tech Project), Indian Institute of Technology Tirupati and Information Sciences Institute
Advisor: [Prof. Jose-Luis Ambite](#)
- Worked on Denoising Diffusion Probabilistic Models (DDPMs) and Generative Adversarial Networks (GANs) to generate brain MRIs conditioned on diagnostic categories: Cognitively Normal, Mild Cognitive Impairment and Dementia.
 - Investigated the correlation between scalar maps of diffusion-weighted MRIs and Alzheimer's disease classification accuracy, with model hyperparameters optimized using Bayesian Optimization.

05/23 – 07/23 **Research Intern**, IUSSTF-Viterbi Summer Internship 2023

Advisor: [Prof. Jose-Luis Ambite](#)

- Developed multi-modal neural networks for brain age prediction (regression) and Alzheimer’s disease diagnosis (classification).
- Analyzed and preprocessed T1-weighted 3D brain MRIs, scalar maps of Diffusion weighted 3D brain MRIs and structured clinical data collected from the ADNI dataset.

ACADEMIC SERVICES

Teaching Assistant

Spring 2026 **DSCI 501** – Principles of Programming for Data Science

Spring 2025 **CSCI 103** – Introduction to Programming

Reviewer

2026 Investigative Ophthalmology & Visual Science (IOVS)

2026 Ophthalmology Science, Elsevier Editorial

ACHIEVEMENTS

2024 **Undergraduate Research Thesis Award** – Graded ‘S’, the highest distinction, for the Undergraduate Research Thesis in Computer Science, IIT Tirupati.

2023 **IUSSTF-Viterbi Scholarship** – One among 15 students selected from India by the Department of Science & Technology, Govt. of India, for pursuing summer research at the USC Viterbi School of Engineering.

SKILLS

Programming Languages – Python, C, C++, Java, Assembly, Verilog

Frameworks/ Libraries – Tensorflow, PyTorch, Pandas, NumPy, SimpleITK, OpenCV

Tools – Git, Anaconda, Docker